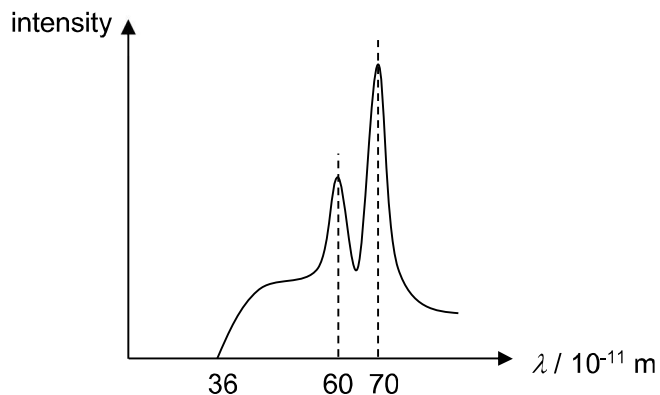


2
9

The x-ray spectrum of a metal target is shown in the figure below.



Which of the following statements is correct?

- | | |
|----------|---|
| A | The smallest wavelength detected, $36 \times 10^{-11} \text{ m}$, is dependent on the target material. |
| B | The graph shows that electrons with a range of kinetic energies are used to bombard the target. |
| C | The locations of the peaks can be used to identify the element that the target material is made of. |
| D | The position of the peaks allow us to calculate the energy of the electrons used to bombard the target. |

Answer: C

Option A is not correct because minimum wavelength (cut-off wavelength) depends on the maximum energy of the incident electrons and independent of the target material.

Option B is not correct because since the X-ray spectra will always contain bremsstrahlung radiation whether the electrons bombarding the metal target has a single energy or have a range of energies.

Option D is not correct because the position (wavelength) corresponding to the characteristic peaks is dependent only on the kind of target and not on the energy of the incoming electrons.

